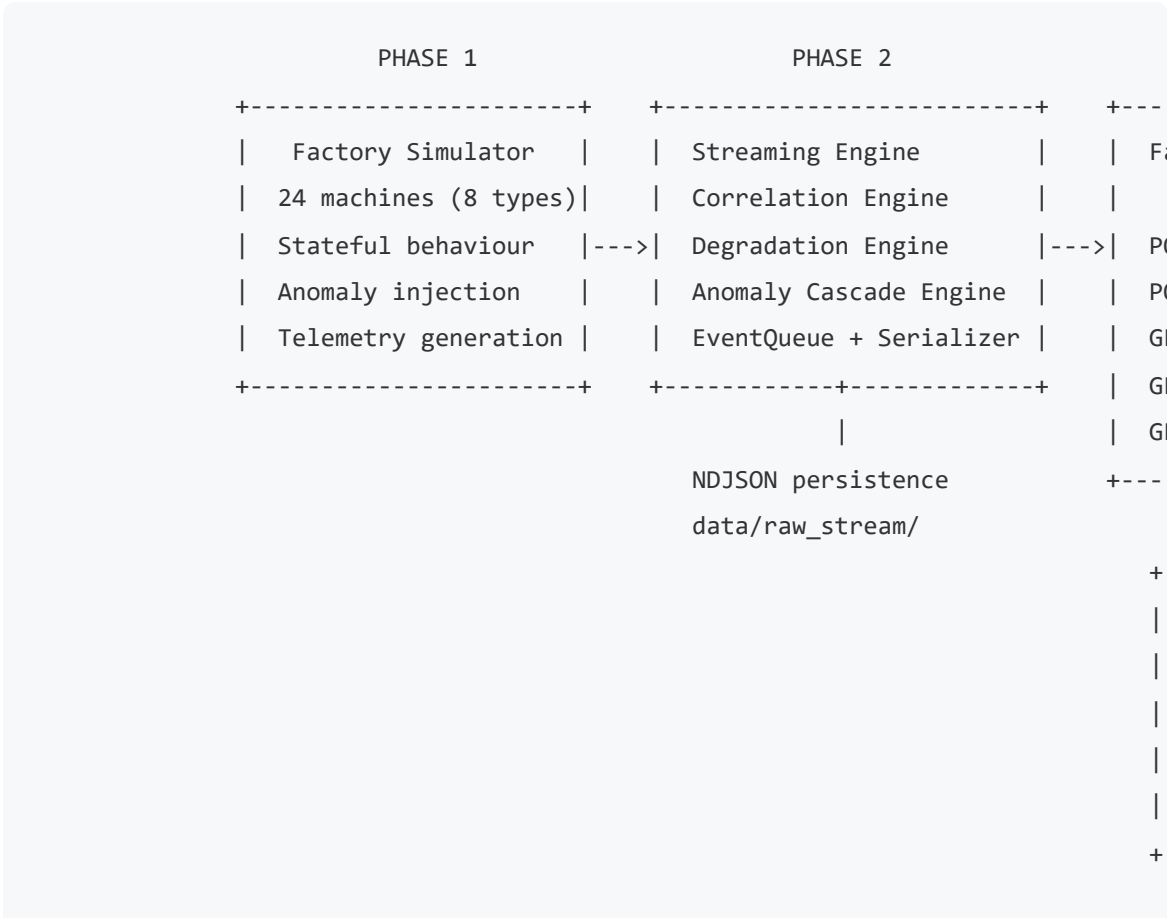


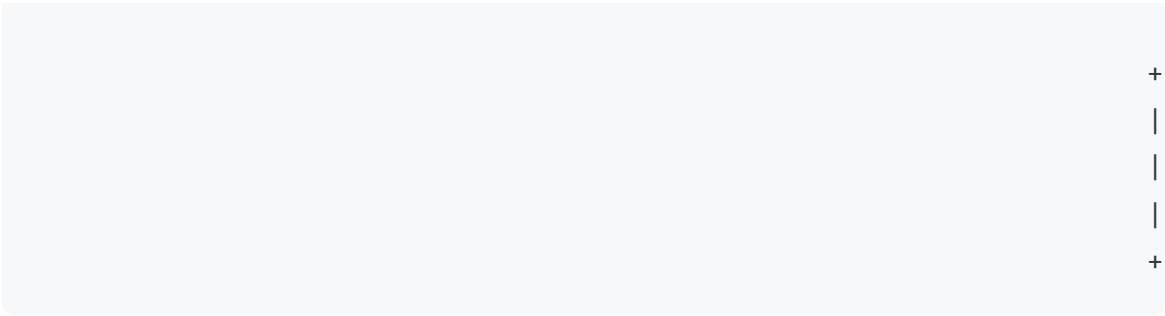
AutoForge Smart Manufacturing Plant

Industry 4.0 Industrial Intelligence Platform -- Phases 1, 2 & 3

A cloud-native real-time telemetry streaming platform that simulates a realistic 24-machine automotive manufacturing plant. The system generates stateful, correlated, anomaly-capable industrial telemetry at 1 Hz per machine, and exposes a production-style FastAPI ingestion gateway for telemetry validation, enrichment, buffering, and persistence.

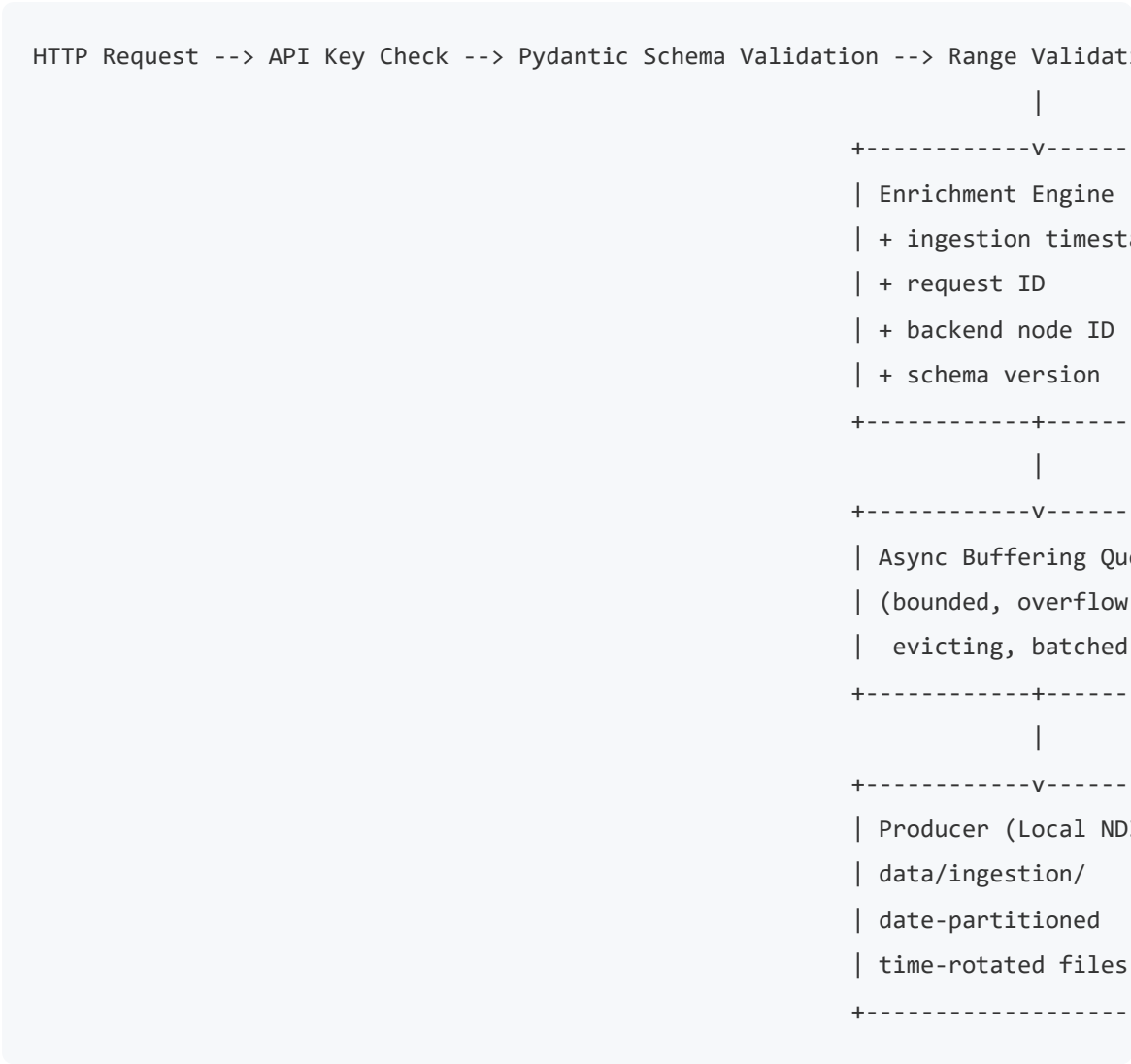
Architecture Overview





Phase 3 -- Backend Ingestion API

Ingestion Flow



API Endpoints

Method	Path	Purpose	Auth
POST	/telemetry	Ingest single telemetry event	Yes
POST	/telemetry/batch	Ingest batch of events	Yes
GET	/health	Service health + uptime	No
GET	/ready	Kubernetes readiness probe	No
GET	/metrics	Operational ingestion metrics	No
GET	/	API root / docs link	No
GET	/docs	Interactive Swagger docs	No

Validation Pipeline

Layer 1 -- Schema Validation (Pydantic v2)

- Required fields, types, nested telemetry structure
- Enum enforcement for `machine_type`, `status`, `anomaly_type`
- Severity/degradation bounds (0.0--1.0)

Layer 2 -- Range Validation (ValidationEngine)

- Per-machine-type telemetry field bounds
- Examples: conveyor RPM 0--5000, CNC tool_wear 0--100%, temperature -10--250 C

Enrichment

Every accepted event receives server-side metadata:

```
{
  "metadata": {
    "ingested_at": "2026-04-26T12:00:01Z",
    "request_id": "req-uuid-here",
    "backend_node": "ingestion-node-01",
```

```
"ingestion_source": "api",
"schema_version": "v1"
}
}
```

Producer Abstraction

```
class ProducerInterface(abc.ABC):
    async def publish_event(event) -> bool
    async def publish_batch(events) -> int
    async def health_check() -> bool
```

LocalProducer -- writes to date-partitioned NDJSON files under `data/ingestion/`. Future **KinesisProducer** will implement the same interface.

Observability

- Structured logging (console + rotating file)
- Request ID middleware (UUID per request)
- Request timing middleware (latency tracking)
- Metrics endpoint: throughput, latency, queue depth, anomaly count, validation failures

API Examples

Ingest a Telemetry Event

```
curl -X POST http://localhost:8000/telemetry \
-H "Content-Type: application/json" \
-H "X-API-Key: autoforge-dev-key-2026" \
-d '{
    "event_id": "d145b4bf-7f7f-4b29-9178-baaeeae48cfa",
    "machine_id": "M008",
    "machine_type": "cnc_machine",
    "factory_id": "AUTOFORGE_01",
```

```
"timestamp": "2026-04-26T12:00:00Z",
"status": "warning",
"telemetry": {
  "spindle_speed": 8042,
  "temperature": 49.3,
  "tool_wear": 52.0,
  "vibration": 2.4
},
"anomaly_detected": true,
"anomaly_type": "excessive_tool_wear",
"anomaly_severity": 0.38,
"degradation_level": 0.12
}'
```

Response (200):

```
{
  "status": "accepted",
  "event_id": "d145b4bf-7f7f-4b29-9178-baaeeae48cfa",
  "ingestion_timestamp": "2026-04-26T12:00:01Z",
  "pipeline_status": "buffered"
}
```

Rejected Event (422)

```
{
  "status": "rejected",
  "event_id": "bad-uuid",
  "errors": [
    {"field": "rpm", "message": "Value 99999 outside range [0, 5000]"},
    {"field": "temperature", "message": "Value -500 outside range [-10, 250]"}
  ]
}
```

Check Health

```
curl http://localhost:8000/health
```

```
{
  "status": "ok",
  "version": "0.3.0",
  "uptime_seconds": 142.5,
  "ingestion_queue_depth": 0,
  "throughput_events_per_sec": 24.0,
  "backend_node": "ingestion-node-01"
}
```

Check Metrics

```
curl http://localhost:8000/metrics
```

```
{
  "total_events_ingested": 3420,
  "events_per_second": 24.0,
  "failed_validations": 3,
  "active_machine_ids": 24,
  "queue_depth": 0,
  "anomaly_event_count": 128,
  "avg_ingestion_latency_ms": 0.071
}
```

Machine Categories

#	Machine Type	Telemetry Fields	Anomalies
4	Conveyor Motor	rpm, temperature, power_consumption	overheating, unstable_rpm
3	Hydraulic Press	hydraulic_pressure, temperature, cycle_time	pressure_drop, overheating

#	Machine Type	Telemetry Fields	Anomalies
3	CNC Machine	spindle_speed, temperature, tool_wear, vibration	excessive_tool_wear, vibration
3	Robotic Arm	joint_load, movement_delay, motor_temperature, accuracy	motor_overheating, delay_spike
2	Industrial Turbine	rpm, vibration, temperature, energy_output	unstable_rpm, vibration
3	Cooling System	coolant_flow_rate, temperature, pressure	pressure_fluctuation, coolant_loss
3	Welding Unit	arc_stability, temperature, energy_usage	unstable_arc, energy_spike
3	Assembly Robot	task_completion_rate, alignment_accuracy, cycle_efficiency, temperature	alignment_degradation, efficiency_drop

Total: 24 machines

Project Structure

```
autoforge/  
├─ backend/                                # Phase 3 -- Ingestion API  
│   └─ api/  
│       └─ routes/  
│           └─ telemetry.py                # POST /telemetry, /telemetry/batch  
│           └─ health.py                   # GET /health, /ready  
│           └─ metrics.py                  # GET /metrics
```

```

| | └─ dependencies.py          # DI container
| └─ core/
| | └─ config.py               # Env-based settings
| | └─ constants.py           # Valid types, ranges
| | └─ logger.py               # Structured logging
| | └─ security.py             # API key validation
| └─ ingestion/
| | └─ ingestion_service.py    # Pipeline orchestrator
| | └─ validation_engine.py    # Range validation
| | └─ enrichment_engine.py    # Metadata stamping
| | └─ producer_interface.py   # Abstract producer
| | └─ local_producer.py       # NDJSON persistence
| | └─ buffering_engine.py     # Async bounded queue
| | └─ retry_engine.py         # Exponential backoff
| └─ models/
| | └─ telemetry_models.py     # Pydantic v2 event models
| | └─ response_models.py      # API response schemas
| | └─ validation_models.py    # Internal validation result
| └─ middleware/
| | └─ request_id.py           # UUID per request
| | └─ logging_middleware.py   # Request timing
| | └─ exception_handlers.py   # Global error handling
| └─ services/
| | └─ telemetry_service.py    # Telemetry route facade
| | └─ health_service.py       # Health data provider
| | └─ metrics_service.py      # Metrics aggregator
| └─ tests/
| | └─ test_backend.py         # 19 tests
| └─ main.py                   # FastAPI app entry point
| └─ requirements.txt
└─ simulator/                  # Phases 1 & 2
| └─ machines/                 # 8 machine types
| └─ telemetry/                 # Correlation, degradation, anomaly, met
| └─ streaming/                 # Streaming engine, event queue, seriali
| └─ storage/                   # NDJSON writer, rotation, replay loader
| └─ factory/                   # Factory simulator orchestrator
| └─ utils/                     # Config, logging, helpers
| └─ main.py                    # Simulator entry point
└─ tests/                       # Phase 1+2 tests (30 tests)
└─ data/
| └─ raw_stream/                # Simulator NDJSON output

```



```
| └─ ingestion/                                # Backend NDJSON output
|─ logs/
|─ .env.example
|─ requirements.txt
└─ README.md
```

Setup Instructions

Prerequisites

- Python 3.11+

Install

```
cd autoforge

# Simulator dependencies
pip install -r requirements.txt

# Backend dependencies
pip install -r backend/requirements.txt
```

Configure

```
cp .env.example .env
# Edit .env to customise intervals, API key, etc.
```

Running

Start the Ingestion API

```
uvicorn backend.main:app --reload
```

The API starts on `http://localhost:8000` with interactive Swagger docs at `/docs` .

Start the Factory Simulator

```
python simulator/main.py
```

Emits telemetry to stdout and persists NDJSON to `data/raw_stream/` .

Run All Tests

```
# Phase 1+2 tests
python -m pytest tests/ -v

# Phase 3 backend tests
python -m pytest backend/tests/ -v
```

Expected Terminal Output (Backend)

```
[STARTUP] AutoForge Ingestion API v0.3.0 node=ingestion-node-01
[BUFFER] Drain loop started
POST /telemetry status=200 latency=1.3ms req_id=1b3c3328-...
[INGESTION] Event d145b4bf-... from M008 accepted
[PERSISTENCE] Opened ingestion file: data/ingestion/2026-04-26/ingestion_12-00.1
[VALIDATION] Event test-bad from M001 failed range checks: [rpm: 99999 outside
POST /telemetry status=422 latency=3.0ms req_id=56ab59cb-...
GET /metrics status=200 latency=1.2ms
GET /health status=200 latency=1.1ms
```

Security

- **API Key:** All telemetry endpoints require `X-API-Key` header
- **CORS:** Configurable origins via `CORS_ORIGINS` env var
- **Request IDs:** Every request gets a UUID for traceability

- Default dev key: `autoforge-dev-key-2026` (change in production)

Performance

- **Async throughout:** FastAPI + asyncio event loop, no blocking I/O
 - **Bounded buffer:** Configurable 10K-event queue with overflow eviction
 - **Batched persistence:** Drain loop processes up to 100 events per cycle
 - **Low latency:** ~0.07ms avg ingestion latency per event locally
 - **Target:** 500+ events/sec on standard hardware
-

Future Roadmap

Phase	Description
4	Cloud Streaming: AWS Kinesis producer, S3 data lake
5	Serverless Processing: Lambda consumers, DynamoDB
6	React Dashboard: Live monitoring, anomaly alerts
7	ML Pipeline: Predictive maintenance models